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Improving forecasting accuracy of medium and long-term runoff using artificial neural network based on EEMD decomposition

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ABSTRACT

Hydrological time series forecasting is one of the most important applications in modern hydrology, especially for the effective reservoir management. In this research, an artificial neural network (ANN) model coupled with the ensemble empirical mode decomposition (EEMD) is presented for forecasting medium and long-term runoff time series. First, the original runoff time series is decomposed into a finite and often small number of intrinsic mode functions (IMFs) and a residual series using EEMD technique for attaining deeper insight into the data characteristics. Then all IMF components and residue are predicted, respectively, through appropriate ANN models. Finally, the forecasted results of the modeled IMFs and residual series are summed to formulate an ensemble forecast for the original annual runoff series. Two annual reservoir runoff time series from Biuliuhe and Mopanshan in China, are investigated using the developed model based on four performance evaluation measures (RMSE, MAPE, R and NSEC). The results obtained in this work indicate that EEMD can effectively enhance forecasting accuracy and the proposed EEMD-ANN model can attain significant improvement over ANN approach in medium and long-term runoff time series forecasting.

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1. Introduction

One of the important tasks of hydrologists and water resource engineers is to assess and forecast the quantity of water available in a basin over longer periods, for example, months and years, and manage the water resource for many practical applications involving conservation, environmental disposal and efficient water supply (Wang et al., 2013). For such purposes, understanding runoff processes at longer time scales and constructing a long-term runoff forecasting model, for example, are much more important than those at smaller time scales (daily or hourly) (Yang et al., 2005). In the past few decades, mathematical modelling of runoff series, for reproducing the underlying stochastic structure of this type of hydrological process, has been performed extensively (Remesan et al., 2009). For example, the Box-Jenkins time series analysis method comprises auto-regressive (AR), moving average (MA), autoregressive moving average (ARMA), etc. (Box et al., 1994). In recent years, non-linear data-driven models have been introduced and widely used as surrogate in hydrological studies as powerful alternative modelling tools, such as artificial neural network (ANN) (Budu, 2014; Chua and Wong, 2011; Dawson et al., 2002; Dibike and Solomatine, 2001; Lu et al., 2004; Minns and Hall, 2006; Wang et al., 2009). A comprehensive review by ASCE Task Committee on Application of Artificial Neural Networks in Hydrology shows the acceptance of ANN technique among hydrologists (Govindaraju and Artific, 2000a, 2000b). ANN has certain advantages in practical applications. For example, ANN models require less information than physically-based models (Sohail et al., 2008a), and they are not like physically-based models which are usually more complex, relying on the skill and experience of the modeler in model calibration (de Vos and Rientjes, 2005). Therefore, the use of ANNs is attractive especially if the interest is solely on making accurate runoff forecasts at a particular location and if the only data available are the time series of flow (Pulido-Calvo and Portela, 2007).

However, despite the good performance of ANN technique when used on their own, there is still room for further improving their accuracy. One new trend is to extract trends and harmonics and eliminate noise from hydrological time series using appropriate data preprocessing techniques (Wu et al., 2010). Hu et al. (2007) developed a rainfall-runoff neural network (RRNN) model using principal component analysis (PCA) as an input data-preprocessing tool, which was found to provide a generally better

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representation of the rainfall-runoff relationship. Wu et al. (2009) used three data-preprocessing techniques, namely, moving average (MA), singular spectrum analysis (SSA), and wavelet multi-resolution analysis (WMRA), to couple with ANN to improve the estimate of daily flows. Sang et al. (2009) developed wavelet analysis (WA) and maximum entropy spectral analysis (MESA) to identify periods of hydrologic series data. Wu and Chau (2011) employed ANN coupled with SSA for rainfall-runoff modeling. Kisi (2009) proposed the application of a conjunction (neuro-wavelet) for forecasting monthly lake levels, and the results indicated that the suggested model could significantly increase the short and long-term forecast accuracy. Sang et al. (2013) used the improved continuous wavelet transform (CWT) method to reveal the periodic characteristics of several typical hydrological series. Wang et al. (2014) developed sample entropy-based adaptive wavelet de-noising approach for meteorologic and hydrologic time series. Nourani et al. (2013) used wavelet transform to extract dynamic and multi-scale features of the nonstationary runoff time series and to remove noise in nerual network based rainfall-runoff modeling. Wei et al. (2013) developed a wavelet-neural network (WNN) hybrid modelling approach for monthly river flow estimation and prediction.

Recently, an ensemble empirical mode decomposition (EEMD) has been developed by Wu and Huang (2009), which is a new noise assisted data analysis method, and which can overcome the mode mixing drawback of the original empirical mode decomposition (EMD), first introduced by (Huang et al., 1998). EEMD method is different from other traditional decomposition techniques such as the Fourier decomposition and wavelet decomposition, and is an empirical, intuitive, direct and self-adaptive data processing method created especially for non-linear and non-stationary signal sequences (Hu et al., 2013; Huang and Wu, 2008). The hydrologic times series (rainfall processes, runoff processes) have characteristics of nonlinear and non-stationary (Karthikeyan and Nagesh Kumar, 2013). Hence, EMD method can be used to analyze the nonlinear data of hydrology. Several successful applications have been reported in the literature that addressed the use of EMD or EEMD to hydrologic time series. Napolitano et al. (2011) discussed the aspects of artificial neural network in hindcasting of daily stream flow data through EMD. Sang et al. (2012) developed empirical mode decomposition (EMD) and maximum entropy spectral analysis (MESA) in combination to identify periods in hydrologic time series. Motivated by the idea of "decomposition and ensemble", the original time series can be decomposed into several sub-series, and each sub-series can be forecasted with the purpose of easy prediction tasks, and the final forecasted value can be obtained by summing the forecasted value of each sub-series (Guo et al., 2012; Wang et al., 2013). EEMD has been employed for decomposing rainfall series in rainfall-runoff model based on support vector machine (SVM) (Wang et al., 2013). Di et al. (2014) proposed a new method with four stages, EMD-EEMD-RBFNN-LNN for predicting the hydrological time series, and the results of six cases show that the proposed hybrid prediction model improves the prediction performance significantly and outperforms some other popular forecasting methods.

The purpose of this paper is to assess forecasting accuracy of artificial neural network (ANN) models coupled with ensemble empirical mode decomposition (EEMD) for annual runoff time series. Hence, EEMD is applied to decompose annual runoff time series, then different ANN models are constructed with each sub-series, and the final forecasted value can be obtained by conjunction of these models. The developed models were evaluated using goodness-of-fit statistics and visual comparison of the hydrograph predictions against actual values. To ensure wider applications of conclusions, two reservoirs annual runoff time series from Biuliuhe and Mopanshan in China are investigated.

The rest of the paper is organized as follows: Section 2 introduces the study areas and the background information. Section 3 gives a brief description to basic theories and algorithms of EEMD, ANN and hybrid EEMD-ANN. The model evaluation criteria of forecasting performance are presented in Section 4. The model development, application results, analysis and discussion are described in Section 5. Section 6 states the conclusions.

2. Study areas and background information

Two case studies are selected for the present model application. The first study area is Biliuhe River, which originates from Qipan Mountain in Liaoning Province, China, passes through Gaizhou, Zhuanghe and Pulandian, and ingresses into Yellow Sea near Xiejiatun. The river catchment approximately covers 2,184 km^2, and the river is about 156 km long. The average annual rainfall of Biliuhe catchment is 742.8 mm. The annual runoff data for this study is from 1951 to 2007 at the dam site of Biliuhe reservoir (Zhang et al., 2011), which has a capacity of 934 million cubic meters and supplies 60% of municipal water of Dalian located in northeast part of Liaoning Province (Zhang et al., 2011). The annual runoff data set from 1951 to 1995 are employed for training models whilst those from 1996 to 2007 are employed for validating performances of model (Fig. 1). According to the runoff time series from 1951 to 2007, the maximum annual runoff is 1.466 billion m^3 occurring in 1964; the minimum annual runoff is 0.109 billion m^3 occurring in 2002; the average annual runoff is 0.608 billion m^3.

The second study area is Lalin River, which is a tributary of Songhuajiang River and originates from Changbai Mountain in Heilongjiang Province in northeast China. The river is about 244 km long, passes through Wuchang, Shulan, Yushu, and ingresses into Songhuajiang River near Shuangcheng. Its catchment approximately covers 1151 km^2 at the Mopanshan dam site of Mopanshan reservoir, which has a capacity of 0.523 billion cubic meters. The average annual rainfall of catchment is about 750 mm. The annual runoff data from 1952 to 2003 are studied at the dam site of Mopanshan reservoir (Li and Chen, 2008), and the data set from 1952 to 1991 are employed for training models whilst those from 1992 to 2003 are employed for validating performances of model (Fig. 2). According to the runoff time series from 1952 to 2003, the maximum annual runoff is 0.903 billion m^3 occurring in 1960; the minimum annual runoff is 0.281 billion m^3 occurring in1978; the average annual runoff is 0.559 billion m^3.

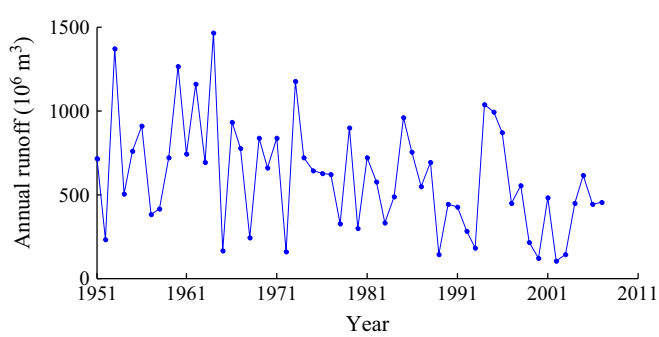

Fig. 1. The annual runoff time series in Biliuhe reservoir.

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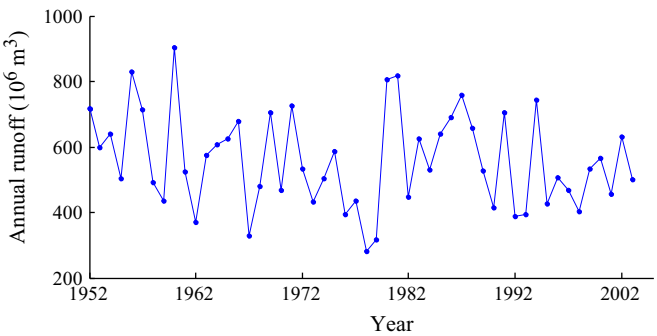

Fig. 2. The annual runoff time series in Mopanshan reservoir.

3. Methodology

3.1. Ensemble empirical mode decomposition (EEMD)

Ensemble empirical mode decomposition (EEMD) (Wu and Huang, 2009) is an enhancement of the empirical mode decomposition (EMD), which is an empirical but highly efficient and adaptive method for processing non-linear and non-stationary time series (Huang et al., 1998; Huang and Wu, 2008). The major idea of EMD is to decompose the original time series data into a finite and small number of oscillatory modes based on the local characteristic time scale (Huang et al., 1998). Each oscillatory mode can be expressed by an intrinsic mode function (IMF) that meets two conditions: first, the functions have the same numbers of extrema and zero-crossings or differ at the most by one; second, the functions are symmetric with respect to local zero mean. The main advantage of EMD is that it does not require *a priori* knowledge and the IMFs can be derived directly from the data itself depending on the data-driven mechanism. The essence of EMD algorithm is a shifting process which extracts the IMF modes from a given time series through (Huang et al. 1998). It can be briefly described as follows:

Step 1. Identify all local extreme including maxima and minima values for a given time series $x(t)$;
Step 2. Connect all local maxima values and all minima values with spline interpolation, respectively, to form an upper envelope $e_{max}(t)$ and a lower envelope $e_{min}(t)$;
Step 3. Use the upper envelope $e_{max}(t)$ and the lower envelope $e_{min}(t)$ to compute the mean $m(t)$ between two envelopes;

$$m(t) = (e_{max}(t) + e_{min}(t))/2 \tag{1}$$

Step 4. Evaluate the difference between the difference of $x(t)$ and $m(t)$ as $h(t)$;

$$h(t) = x(t) - m(t) \tag{2}$$

Step 5. Check whether or not $h(t)$ satisfies the two conditions of IMF properties according to stopping criteria. If they are satisfied, $h(t)$ is denoted as the first IMF [written as $c_1(t)$ and 1 is its index]; If $h(t)$ is not an IMF, $x(t)$ is replaced with $h(t)$ and iterate steps 1–4 until $h(t)$ meets the two conditions of IMF properties.
Step 6. The residue $r_1(t)=x(t)-c_1(t)$ is then treated as new data subjected to the same shifting process as described above for the next IMF from $r_1(t)$. Finally, the whole decomposition is completed with a finite number of IMFs until the residual satisfies some stopping criteria.

The stopping criterion presented by Huang et al. (2003) for extracting an IMF is: iterating predefined times after the residue satisfies the restriction that the number of zero-crossings and extrema do not differ by more than one. The shifting process stops when the residue $r(t)$ becomes a monotonic function or at most has one local extreme point from which no more IMF can be extracted. Having determined successively the different IMFs $c_1(t)$, $c_2(t)$,…,$c_n(t)$ and $r_n(t)$ and the original time series can be decomposed into n modes and a residue as follows:

$$x(t) = \sum_{i=1}^{n} c_i(t) + r_n(t) \tag{3}$$

where n is the number of IMFs, $r_n(t)$ denotes the final residue and $c_i(t)$ are nearly orthogonal to each other, and all have zero means.

However, EMD has a potential mode-mixing problem, which can render EMD unable to represent the characteristics of the original data (Wu and Huang, 2009). To overcome the mode-mixing problem, the ensemble empirical mode decomposition (EEMD) has been developed by Wu and Huang (2009), who define the true IMF components as the mean of an ensemble of trials, each consisting of a signal added with a white noise of finite amplitude. The added white noise establishes a uniform reference background in the time-frequency space so that the bits of signals of different scales are automatically projected onto proper scales of reference. The effect of added white noise can be controlled by following the statistical rule (Wu and Huang, 2009):

$$\varepsilon_n = \frac{\varepsilon}{\sqrt{N}} \tag{4}$$

where N is the ensemble number, ε is the amplitude of the added white noise and ε_n is the standard deviation of the error, which is defined as the difference between the input signal and the corresponding IMFs. In this study, the ensemble number is set to 100 and the amplitude of added white noise is set to 0.2 times of standard deviation of that of the data based on suggestion by Wu and Huang (2009). Therefore, based on EMD algorithm (Huang et al., 1998), EEMD method can be briefly described as follows (Wu and Huang, 2009):

Step 1. Set the ensemble number and the amplitude of the added white noise.
Step 2. Add a white noise series to the targeted data with the set amplitude.
Step 3. Decompose the data with added white noise into IMFs.
Step 4. Repeat steps 2 and 3 again and again, but with different white noise series each time, then, the final ensemble means of corresponding IMFs of the decompositions are obtained.

3.2. Artificial neural network (ANN)

Today, ANN is one of the most commonly used artificial intelligence technique being applied in a variety of application with great success. As an information processing system, ANN is composed of many nonlinear and densely interconnected processing elements or neurons, which are arranged in groups called layers (Sudheer et al., 2002). The main advantages of the ANN model are that it does not require information about the complex nature of the underlying process under consideration to be explicitly described in mathematical form (Wang et al., 2009). ANN is a proved and efficient method for modeling complex input–output relationships in hydrologic time series forecasting (Kisi and Kerem Cigizoglu, 2007).

Among the many neural network architectures, the feed forward neural network with back propagation training algorithm, which is the most popular training approach, was used in hydrological modeling (Govindaraju and Artific, 2000a, 2000b). The structure of a typical three-layer feed-forward ANN is shown in

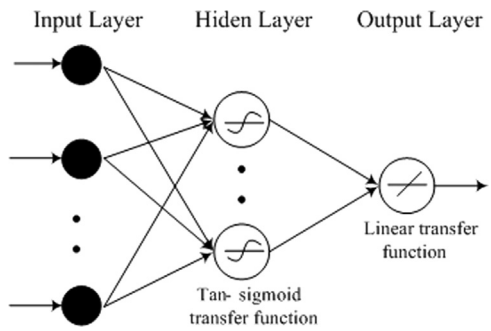

Fig. 3. Architecture of three layer feed-forward back-propagation ANN.

Fig. 3. As it can be seen from Fig. 3, the network contains one input layer, where the data are introduced to the network, one hidden layer with n neurons, where data are processed and one output layer, where the results of given input are produced. Mathematically, the network can be expressed as follow:

$$y = f\left(\sum_{i=1}^{n} w_i x_i + b\right) \tag{5}$$

where y is the output, f is the transfer function, w_i is the weight vector, x_i is the input vector, and b is the bias. Therefore, to use Eq. (2) to runoff predictions, the training algorithm is required to optimize w and b.

In this study, a three-layer feed-forward ANN model trained with Levenberg–Marquardt (LM) algorithm (Asadi et al., 2012) is used, and the tan-sigmoid transfer function is adopted in determining the neurons of the hidden layer whilst the linear transfer function is used in determining the neurons of the output layer. The training epoch is set to 1000. Before the ANN training, the data normalization is also an necessary step for ANN modelling, because this can ensure that all variables receive equal attention and that the efficiency of the training network is improved (Dawson et al., 2002). All the data series are normalized using the minimum ($q_{\min}$) and maximum ($q_{\max}$) values as described in Eq. (3), so that the variable values range from 0 to 1.

$$normalized\ x = \frac{q - q_{\min}}{q_{\max} - q_{\min}} \tag{6}$$

3.3. The hybrid EEMD-ANN forecasting model

Most of the hydrologic data are non-stationary in nature (Milly et al., 2008), especially in medium and long term runoff. This leads to poor generalization and undesirable forecasting performance of many models because they impose a number of pseudo-variation requirements on models and this affects the correct understanding of data variations. Therefore, in order to improve the forecasting accuracy of annual runoff, this paper presents the hybrid EEMD-ANN model. The methodological procedures of the EEMD-ANN forecasting model can be demonstrated in Fig. 4.

As can be seen from Fig. 4, the three main steps of the presented EEMD-ANN forecasting paradigm can be summarized as following:

(1) Firstly, apply the EEMD technique to decompose the original annual runoff time series $x(t)$ (t=1, 2, …, n) into m IMF components, $c_i(t)$, i=1, 2, …, m, and one residual component $r_m(t)$.
(2) Secondly, use the ANN model to build a forecasting model for each extracted IMF component and the residual component, and to make the corresponding prediction for each component, respectively.
(3) Finally, the forecasting results of all extracted IMF and residual components obtained by appropriate ANN model, are combined to generate an aggregated output, which can be used as the final forecasting result for the original annual runoff time series.

To summarize, the hybrid EEMD-ANN forecasting model applies the idea of "decomposition and ensemble". The decomposition is to simplify the forecasting task, while the ensemble is to formulate a consensus forecasting on original runoff data. In order to verify the pattern of the extracted IMFs and residual components to reflect the forecasting model and to enhance the forecasting performance, two annual runoff time series from two river reservoirs, namely, Biuliuhe and Mopanshan in China, are investigated for testing purpose in this study.

4. Model performance evaluation

In order to evaluate the forecasting ability of the developed models, four main criteria, which have been widely used to evaluate the goodness-of-fit of hydrologic and hydro-climatic models (Legates and McCabe, 1999), are employed for evaluation of level prediction and directional forecasting, respectively.

(1) Root mean squared error (RMSE)

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(q_f(i) - q_0(i))^2} \tag{7}$$

where $q_0(i)$ and $q_f(i)$ are, respectively, the observed and forecasted runoffs, and N is the number data points considered.
As one of the commonly used error index statistics, RMSE can provide a good measure of model performance for high flows (Karunanithi et al., 1994).

(2) Mean absolute relative error (MARE)

$$MARE = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{q_f(i) - q_0(i)}{q_0(i)}\right| \times 100 \tag{8}$$

MARE is an unbiased statistic for measuring the predictive capability of a model (Wang et al., 2009).

(3) Coefficient of correlation (R)

$$R = \frac{(1/N)\Sigma_{i=1}^{N}(q_0(i) - \bar{q}_0)(q_f(i) - \bar{q}_f)}{\sqrt{(1/N)\Sigma_{i=1}^{N}(q_0(i) - \bar{q}_0)^2} \times \sqrt{(1/N)\Sigma_{i=1}^{N}(q_f(i) - \bar{q}_f)^2}} \tag{9}$$

where $\bar{q}_0$ and $\bar{q}_f$ are, respectively, the mean observed and forecasted runoffs.
R has been widely used for model evaluation, though they are oversensitive to high extreme values (outliers) and insensitive to additive and proportional differences between model predictions and measured data (Legates and McCabe, 1999; Wang et al., 2009).

(4) Nash-Sutcliffe efficiency coefficient (NSEC)

$$NSEC = 1 - \frac{\Sigma_{i=1}^{N}(q_0(i) - q_f(i))^2}{\Sigma_{i=1}^{N}(q_0(i) - \bar{q}_0)^2} \tag{10}$$

NSEC introduced by Nash and Sutcliffe (Nash and Sutcliffe, 1970) has a range between 1 (best fit) and $-\infty$. An efficiency of lower than zero indicates that the mean value of the observed time series would have been a better predictor than the model (Krause et al., 2005).

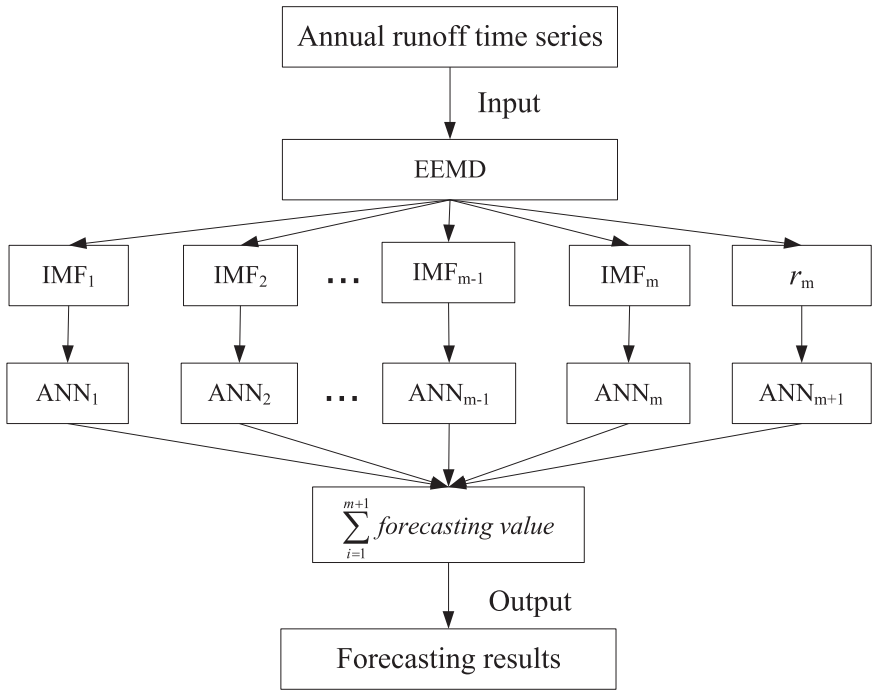

Fig. 4. The architecture of hybrid EEMD-ANN model.

5. Model development and application

5.1. Decomposing annual runoff time series using EEMD

By employing the EEMD technique, the two original annual runoff time series are decomposed into several independent IMFs and one residue, respectively. The results are illustrated in Figs. 5 and 6. As can be seen from Figs. 5 and 6, the two original annual runoff time series are decomposed into four independent IMF components in the order from the highest frequency to the lowest frequency, and one residue component, respectively. The IMF1, IMF2 IMF3 and IMF4 components in Figs. 5 and 6 show that there are obvious periodic variabilities within 3–4 years, 6–7 years, 11 years and 20–22 years, respectively. This is similar to the finding from Pekárová et al. (2003), who analyzed runoff oscillation of the main rivers of the world during the 19th–20th Centuries. Williams (1961) investigated the nature and causes of cyclical changes in hydrological data of the world, and pointed out

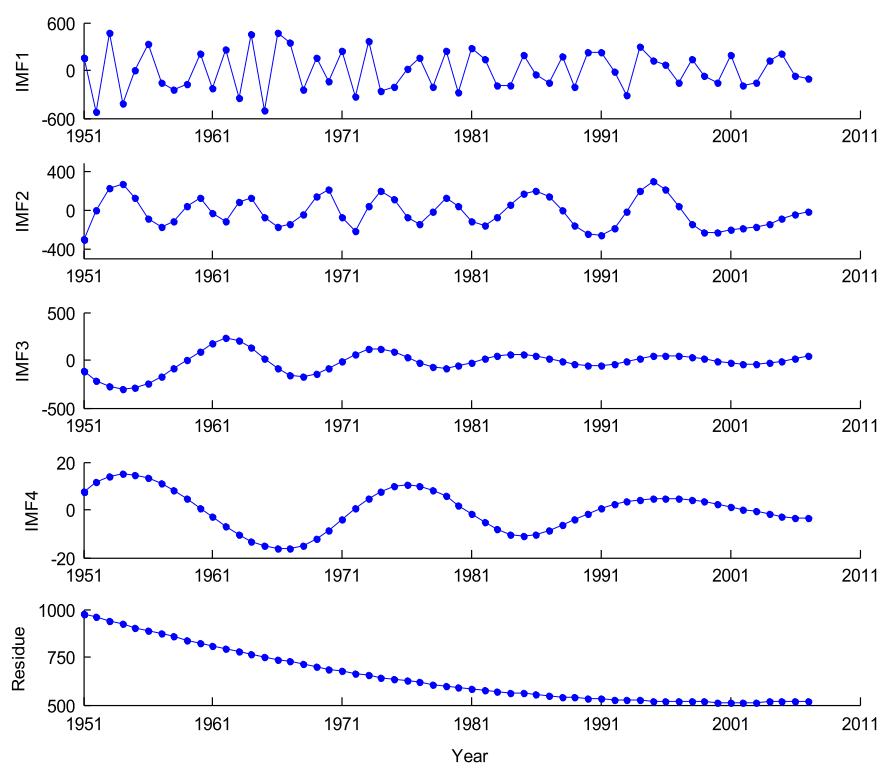

Fig. 5. Decomposition of annual runoff time series in Biliuhe reservoir.

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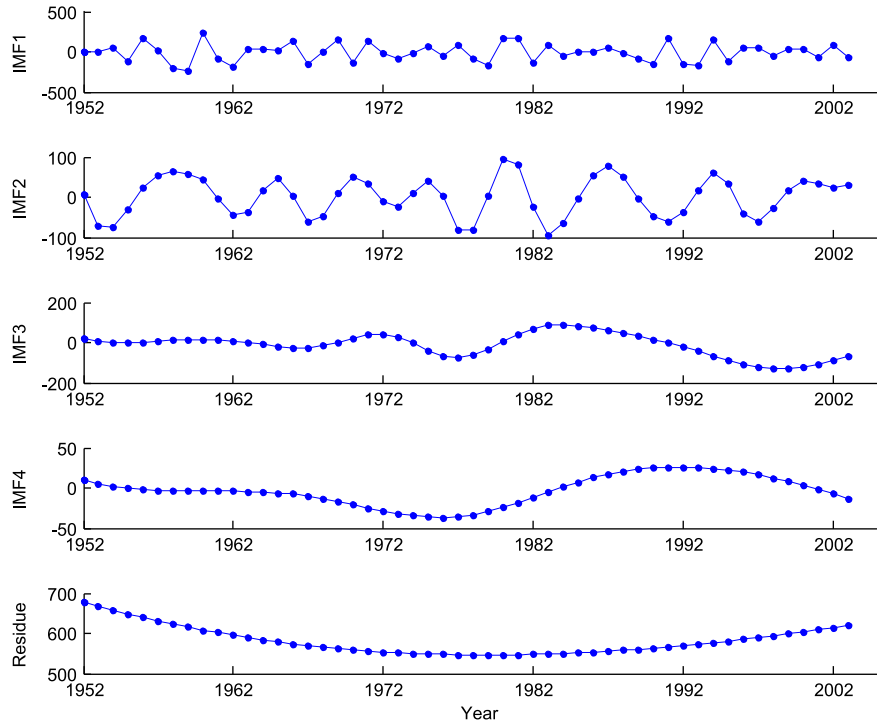

Fig. 6. Decomposition of annual runoff time series in Mopanshan reservoir.

that the most frequently studied cycles in connection with runoff variability are the 11-year (22-year) Hale cycles. The residue component demonstrates the overall trend of annual runoff time series. Therefore, the decomposition can be helpful to transform non-linear and non-stationary time series to stationary time series and can be useful to improve the prediction capacity.

5.2. Modelling approach

When ANN is used to forecast annual runoff time series based on historical observed records, a key problem is how to choose the input variables. Generally empirical relationships between input and output have been used to identify inputs, which were used in hydrological prediction (Kisi and Shiri, 2011). Therefore, several input combinations are tried using ANN to forecast annual runoff in two reservoirs studied in this paper. The antecedent p values ($t-1$, $t-2$,…, $t-p$) are tested to predict a value of current time (t). Therefore, the following combinations of input data of original annual runoff or IMFs and residue component values are evaluated:

$$q_f(t) = ANN(q(t-1), q(t-2)) \tag{11}$$

$$q_f(t) = ANN(q(t-1), q(t-2), q(t-3)) \tag{12}$$

$$q_f(t) = ANN(q(t-1), \ldots, q(t-4)) \tag{13}$$

$$q_f(t) = ANN(q(t-1), \ldots, q(t-5)) \tag{14}$$

$$q_f(t) = ANN(q(t-1), \ldots, q(t-6)) \tag{15}$$

In order to obtain satisfactory prediction accuracy, the number of nodes in the hidden layer needs to be determined. However, there is no rule for specifying the number of nodes in the hidden layer (Sohail et al., 2008b). Most of the researchers selected the number by trial and error method which is also recommended by Shamseldin et al. (2002). Therefore, the numbers of nodes ranging from 2 to 10 were tried in the hidden layer and their errors were compared. So, models having the least number of nodes in the hidden layer with minimum errors were selected in this study.

5.3. Application analysis and discussion

Based on the aforementioned description, the original annual runoff time series and the decomposed IMFs and residue component were modelled. According to performance evaluation measures, the best fitted models have been identified out of the various competing models for the original annual runoff time series and the decomposition of annual runoff time series in the two reservoirs. The obtained ANN models are demonstrated in Table 1.

Four statistical performance evaluation measures, namely, RMSE, MARE, R and NSEC, are employed to evaluate the performances of ANN models based on original runoff time series and the proposed EEMD-ANN models, and the statistical results of different models are summarized in Table 2. From Table 2, it can be observed that ANN based on EEMD decomposed annual runoff series is able to produce a good and close forecast, as compared to the ANN with the original annual runoff time series for the annual

Table 1
Adopted structure of ANN.

Name	Sample data set	p	Numbers of nodes
Biliuhe	Original	5	4
	IMF1	5	4
	IMF2	5	4
	IMF3	5	4
	IMF4	5	4
	Residue	5	4
Mopanshan	Original	5	6
	IMF1	4	10
	IMF2	5	8
	IMF3	5	4
	IMF4	5	4
	Residue	5	4

Table 2
Forecasting performance indices of models for two annual runoff time series.

Name	Model	Training				Validation			
		RMSE	MARE	R	NSEC	RMSE	MARE	R	NSEC
Biliuhe	ANN	191.29	35.18	0.816	0.644	342.85	149.05	0.135	−1.474
	EEMD- ANN	146.01	27.49	0.892	0.793	151.69	61.96	0.726	0.516
Mopanshan	ANN	64.21	8.36	0.907	0.819	123.68	19.83	0.304	−0.489
	EEMD- ANN	45.13	6.19	0.959	0.911	67.56	11.65	0.844	0.556

runoff forecasting in the two reservoirs.

For Biliuhe reservoir, in the training phase, the EEMD-ANN model improved the ANN model with 23.67% and 21.85% reduction in RMSE and MARE, respectively, and improvements of the forecasting results regarding the R and NSEC were approximately 9.31% and 23.14%, respectively. In the validation phase, the EEMD-ANN model improved the ANN forecast with 55.76% and 58.43% reduction in RMSE and MAPE, respectively, and improvements of the forecast results regarding the R and NSEC were 437.78% and 135.01%, respectively.

For Mopanshan reservoir, in the training phase, the EEMD-ANN model improved the ANN model with 29.71% and 25.96% reduction in RMSE and MARE, respectively, and improvements of the forecasting results regarding the R and NSEC were 5.73% and 11.23%, respectively. In the validation phase, the EEMD-ANN model improved the ANN forecast with 45.37% and 41.25% reduction in RMSE and MAPE, respectively, and improvements of the forecast results regarding the R and NSEC were 177.63% and 213.70%, respectively.

Results of this analysis demonstrate that the proposed EEMD-ANN model is able to attain better result than ANN model with drastic improvement in terms of different evaluation measures for annual runoff time series forecasting. Figs. 7–10 illustrate runoff forecasting results using ANN and EEMD-ANN models. It can be seen from Figs. 7–10 that the EEMD-ANN model can mimic runoff better than that by ANN model. This also indicates that the EEMD model is suitable for decomposing annual runoff time series, and the idea of "decomposition and ensemble" is feasible and the proposed EEMD-ANN model can overcome drawbacks of individual models by generating a synergetic effect in forecasting. Therefore, the annual runoff data decomposed by using EEMD technique as input data of models can improve the prediction performance.

6. Conclusions

In order to improve the forecasting accuracy of medium and long-term runoff, this paper proposes a hybrid forecasting model

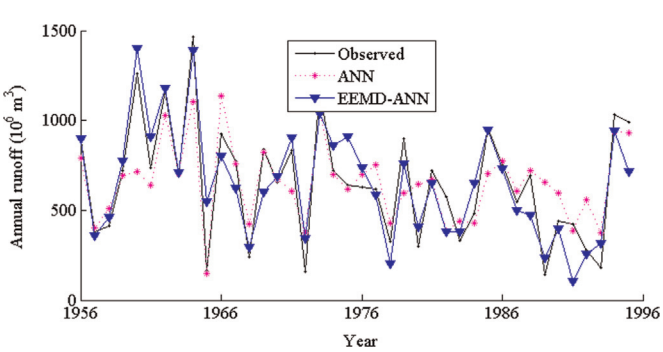

Fig. 7. ANN, EEMD-ANN forecasted and observed runoff during training period in Biuliuhe Reservoir.

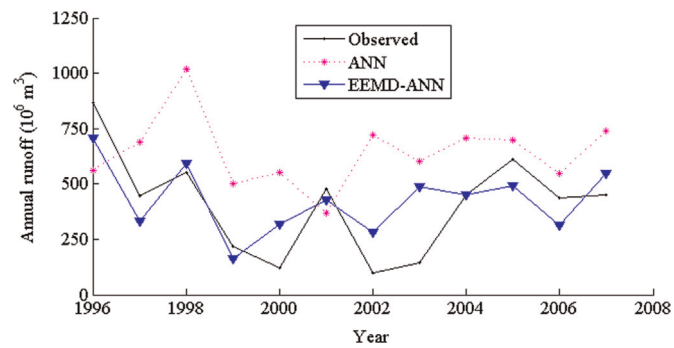

Fig. 8. ANN, EEMD-ANN forecasted and observed runoff during validation period in Biuliuhe Reservoir.

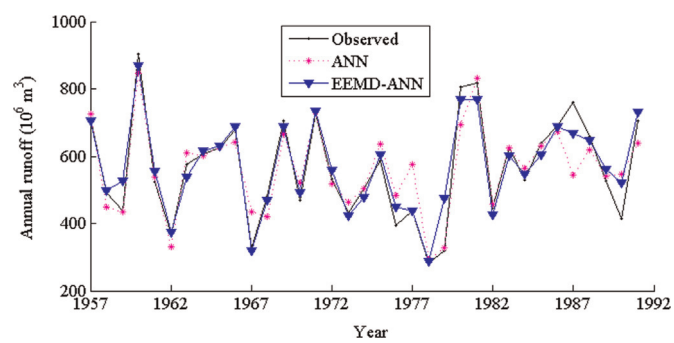

Fig. 9. ANN, EEMD-ANN forecasted and observed runoff during training period in Mopanshan Reservoir.

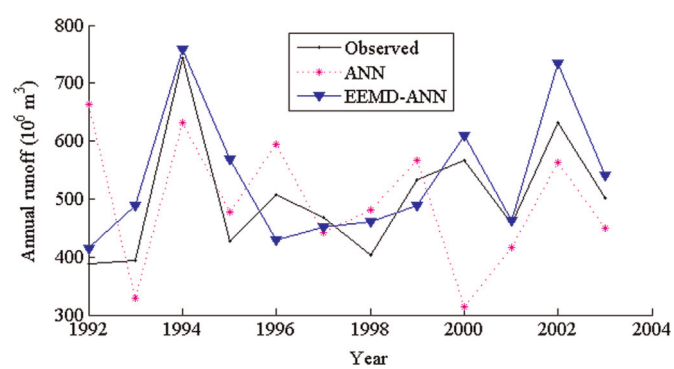

Fig. 10. ANN, EEMD-ANN forecasted and observed runoff during validation period in Mopanshan Reservoir.

based on ensemble empirical mode decomposition (EEMD) and three-layer feed-forward ANN model to forecast the annual runoff time series. An ANN model based on the original annual runoff time series is also employed as a benchmark comparison. Based on annual runoff data from Biuliuhe and Mopanshan in China, the models are developed, and four statistical performance evaluation measures (RMSE, MAPE, R and NSEC) are adopted to evaluate the performances of various models. The results obtained in this work

indicate that EEMD can effectively enhance forecasting accuracy and the proposed EEMD-ANN model can significantly improve ANN model for annual runoff time series forecasting. Thus, applying this hybrid method to forecast annual runoff is very important for future studies. Furthermore, there are several advantages of the proposed methodology. Firstly, the basic principle of the EEMD is simple, yet can provide deep insight into the characteristics of annual runoff time series. Secondly, ANN forecasting only requires small amount of runoff data in question. Thirdly, the zero mean of IMF components is helpful for performing ANN modeling. Finally, the proposed models do not entail complicated decision-making about the explicit form in each particular case. Therefore, developing a hybrid forecasting model by incorporating EEMD may lead to more accurate and stable forecasting results, and may also be helpful in studies that are concerned with hydrological time series forecasting for a wide range of problems related to effective reservoir management.

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